Crypto Cash-and-Carry: Capital Efficiency in the Spot–Perpetual Basis Trade

Summary

Question. Buying spot crypto and shorting the perpetual future ("perp") collects the perp's funding rate with no price exposure: economically, a secured loan of cash. Can the trade be improved by arbitraging funding across exchanges, by forecasting funding, or by optimising which coins to hold?

Approach. Free exchange data for 274 coins on Binance and Bybit (2021–2026, including later-delisted coins), plus two on-chain venues (Hyperliquid, dYdX) from 2023. Every test had its decision rule written down before it ran, and every variant is counted when judging statistical significance.

Findings.

- **Cross-venue arbitrage does not pay.** Binance and Bybit funding is 0.95 correlated; gaps close within hours and rarely exceed trading costs. Gaps to an on-chain venue are about twice as wide.
- **Carry over cash collapsed after 2021.** The exchange's funding floor (10.95% a year) stayed fixed while T-bills rose from 0% to 5%.
- **Funding is forecastable, but the edge is about the size of trading costs.**
- **No active strategy beat holding BTC and ETH passively**, including a daily MILP optimiser.
- **Capital structure decides the outcome.** The trade breaks even at about 1.4× capital per $1 of notional. Under portfolio margin (about 1.05×), passive BTC+ETH earned **+1.9% a year over T-bills (Sharpe 4.2)**; with spot and margin held separately (1.5×), +0.4%.

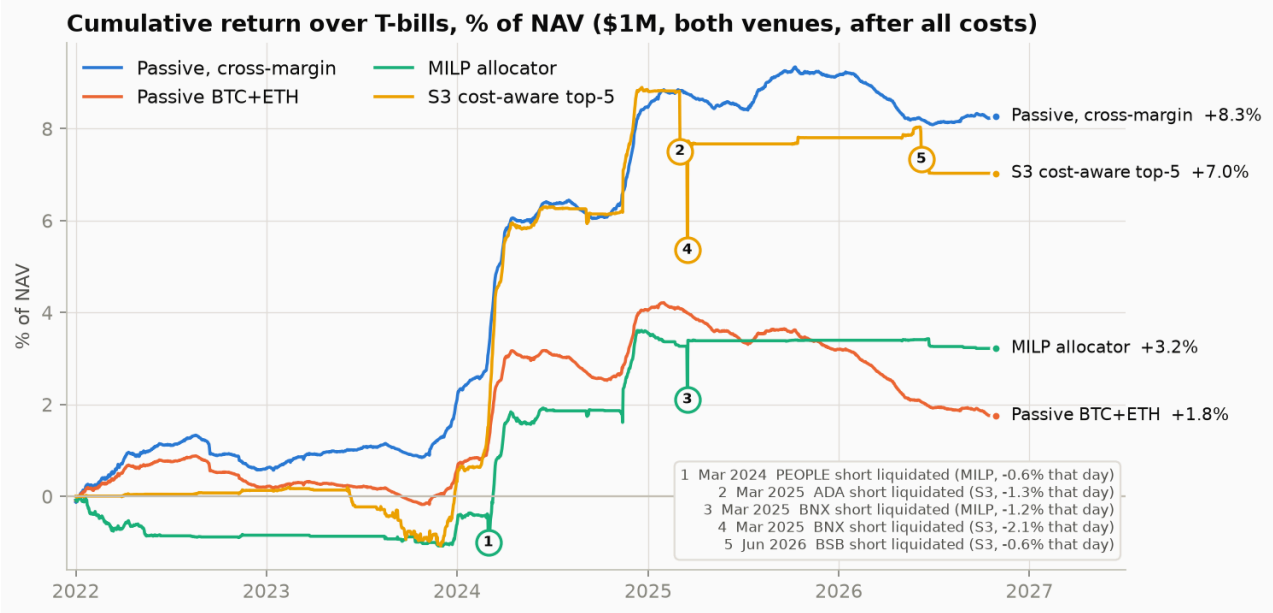


Figure 1. Cumulative return over T-bills, $1M across both exchanges, after all costs (2022–2026). Labelled dips are short-perp liquidations in price spikes on small coins (Section 3.3).

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1. How the trade works

Perpetual futures never expire. Instead, longs and shorts exchange a **funding payment** every 1–8 hours that pushes the perp back towards the spot price:

- Perp above spot → longs pay shorts. Perp below spot → shorts pay longs.
- Binance's formula per 8 hours: **funding = premium + clamp(0.01% − premium, ±0.05%)**.
- For any small premium, the clamp returns exactly **0.01% per 8 hours = 10.95% a year**: an administered **floor**, much like an interest rate set by the exchange.

The basis (cash-and-carry) trade: buy q coins of spot, short q coins of the perp.

P&L component	Source
Funding	received on the short perp each period (paid when funding is negative)
Basis change	gain or loss as the perp–spot gap moves
Trading costs	fees and slippage when opening, closing or resizing
Cost of capital	the T-bill interest given up on cash tied up in the position

Margin factor = capital tied up per $1 of notional (spot plus margin for the short):

- **1.5** if spot and the short perp sit in separate accounts with a 50% buffer (the conservative case).
- **About 1.05** under portfolio margin, where spot counts as collateral at a 95% ratio.

Excess return over T-bills ≈ funding ÷ margin factor − T-bill rate − trading costs.

All results below are in excess of the 3-month T-bill, net of all costs.

2. Data

- **Exchanges:** Binance and Bybit (funding, daily volume, 8-hour spot and perp prices, 2021–2026); Hyperliquid and dYdX funding from 2023; 3-month T-bill from FRED.
- **Universe:** each month, the top 30 coins by trailing Binance volume among coins on both exchanges, using only data available at the time: **274 coins**, including coins later delisted.
- **Investable:** in that month's universe, with a spot market on the same exchange and a perp–spot gap within ±20%. About **25 coins per exchange** from 2022.
- Stock, ETF and commodity perps are excluded. Full construction rules: Appendix A.

3. What was tried

3.1 Cross-exchange arbitrage

Idea. If funding differs between exchanges, short the perp where funding is high and buy it where it is low. The position is price-neutral and collects the gap.

Test (decision rule fixed in advance):

- Signal: the average Binance–Bybit funding gap over the previous 3 days.
- Outcome: the gap collected over the next 7 days, in the signal's direction.
- Pass: the top 20% of signals must beat the 21 bp round-trip cost on average **and** in at least 4 of 6 years.

Result: fails. The top 20% collected 25 bp on average but beat costs in only 3 of 6 years.

	2021	2022	2023	2024	2025	2026
Funding correlation, Binance vs Bybit	0.71	0.82	0.95	0.95	0.94	0.95
Top-20% gap collected over 7 days (bp)	55.8	17.6	21.1	21.0	20.3	9.8

Why. Arbitrageurs already keep the two largest exchanges in line: the typical gap is 0.2–0.7 bp per 8 hours and halves in about 4.5 hours. What is left sits inside the cost of trading it.

Where moving money is harder, gaps are wider. On Hyperliquid, an on-chain exchange where capital moves by bridging stablecoins rather than an exchange withdrawal, gaps against Binance on the same coins are:

	Binance–Hyperliquid	Binance–Bybit (same coins)
Median gap (bp per 8 hours)	0.62	0.32
90th-percentile gap	3.40	1.63
Top-20% gap collected over 7 days	46 bp	13 bp
Share of those trades beating the round-trip fee	69%	24%

- The largest Hyperliquid gaps beat trading fees most of the time, but only before bridging costs, transfer delays and on-chain risk, which is exactly what keeps them open.
- The gap is shrinking: the top-20% 7-day gap fell from 54 bp (2024) to 14 bp (2026).
- dYdX shows wider gaps too, but its funding formula has no floor, so its gaps are largely mechanical and are not counted (Appendix A).

3.2 Forecasting funding

Setup (details in Appendix B):

- Target: each coin's average funding over the **next 7 days**.

- Four models: a constant (the 0.01% floor), last value, an exponentially weighted average (EWMA) and a pooled linear regression on recent funding averages, the perp–spot gap and time spent at the floor.
- Walk-forward: trained on 2021, tested out of sample from 2022, refitted monthly using only past data.

Results (out of sample):

	Floor	Last value	EWMA	Regression
Forecast error, mean absolute (bp per 8h), range over 2022–26	1.06–2.17	0.96–2.00	0.82–1.68	**0.72–1.39**
Ranking of coins: rank IC, range over years	—	0.37–0.60	**0.49–0.70**	0.47–0.68
Top 20% by forecast vs average coin (% a year, before costs)	—	4.6–10.5	6.5–15.2	6.3–16.8

- Funding is clearly forecastable: a rank IC around 0.5 is very high (Appendix B explains the measure).
- The regression wins on error in 4 of 5 years and is used downstream; the EWMA ranks coins slightly better.
- **The catch:** rotating weekly into the top 20% costs about 15.6% a year (0.38% round trip × 52), about the same as the extra funding it identifies.

3.3 Rules for choosing coins

Strategies ($1M split across both exchanges, margin factor 1.5, 2022–2026):

	Rule
Passive	hold BTC and ETH on both exchanges, always
Naive top 5	each week, rotate into the 5 coins with the highest forecast
Cost-aware top 5	enter a coin only if its forecast beats T-bills plus the round-trip cost spread over 7 days; exit only when the forecast falls below T-bills

Results:

	Excess return a year	Sharpe	Max drawdown	Turnover (× capital a year)
Passive BTC+ETH	**+0.36%**	**1.10**	−2.5%	0.1
Naive top 5	−4.31%	−1.68	−19.0%	31.3
Cost-aware top 5	+1.46%	0.83	−3.5%	3.0
Cost-aware top 5, EWMA forecast	+2.56%	1.35	−2.3%	2.4

None of the active rules beats passive on a risk-adjusted basis (the cost-aware rule's Sharpe is lower, P(not better) = 0.51), and none survives the correction for the number of variants tried (Appendix B).

Why:

- **Naive rotation is eaten by costs:** the most funding collected (5.8% of capital a year) but 6.0% paid in costs and 0.6% in liquidation penalties.
- **The cost-aware rule mostly sits in cash:** spreading a 0.38% round trip over only 7 days requires forecast funding of about 24% a year to enter, so it held 1.7 coins on average out of 10 slots.
- **Small coins bring liquidations.** Every numbered dip in Figure 1 is a short perp liquidated in a price spike on a small coin: PEOPLE (March 2024), ADA (March 2025, when it jumped on the US "crypto strategic reserve" announcement), BNX (March 2025, −2.1% in a day, mostly reversed the next day) and BSB (June 2026). Each costs a 2% penalty plus a loss on the leftover spot, which is sold at the next 8-hour close after the spike has partly reversed. BTC and ETH were never liquidated.

3.4 A daily optimiser (MILP)

How it is used. Once a day, a mixed-integer program (solved with HiGHS) decides how much of each coin to hold on each exchange, and whether to move capital between exchanges. Trades execute at the next 8-hour close.

- **Maximises:** expected funding over 30 days, minus the T-bill on capital used, minus trading costs (charged in full when a position opens), minus a risk penalty for small and newly listed coins.
- **Constraints:** capital per exchange; per-coin caps (25% of an exchange's capital for BTC/ETH, 10% other coins, 5% coins listed under 90 days); at most 1% of a coin's daily volume; $10k minimum position; at most 12 positions; at most 60% of capital on one exchange; transfers of at least $50k that arrive a day later.
- **Forecast:** the regression, refitted for a 30-day target.
- Full formulation: Appendix D.

Results:

	Margin	Excess return a year	Sharpe	Turnover	Costs a year
Passive BTC+ETH	1.50	+0.36%	1.10	0.1	0.03%
MILP	1.50	+0.67%	0.67	5.3	1.07%
MILP with turnover controls*	1.05 / 1.30	+1.53%	1.32	4.1	0.82%
Passive BTC+ETH, portfolio margin	**1.05**	**+1.94%**	**4.18**	0.2	0.04%

** Exit costs charged in the objective, a no-trade band (trades smaller than $5k or 10% skipped), and realistic portfolio margin (1.05 for BTC/ETH, 1.30 for other coins).*

Why it does not beat passive:

- It does what it was designed to: fully invested during the high-funding periods of 2024, in T-bills otherwise, with fewer liquidations than the coin-picking rules (Figure 2).
- But it **gives much of the carry back in trading:** about 100 opens and closes a year as forecasts change (Figure 3). Turnover controls cut that by a third and doubled the return, still well short of

passive under the same margin.

- The extra funding in small coins comes with **higher margin factors, liquidations and basis losses**. Passive BTC+ETH trades almost never and ties up the least capital (Figure 4).

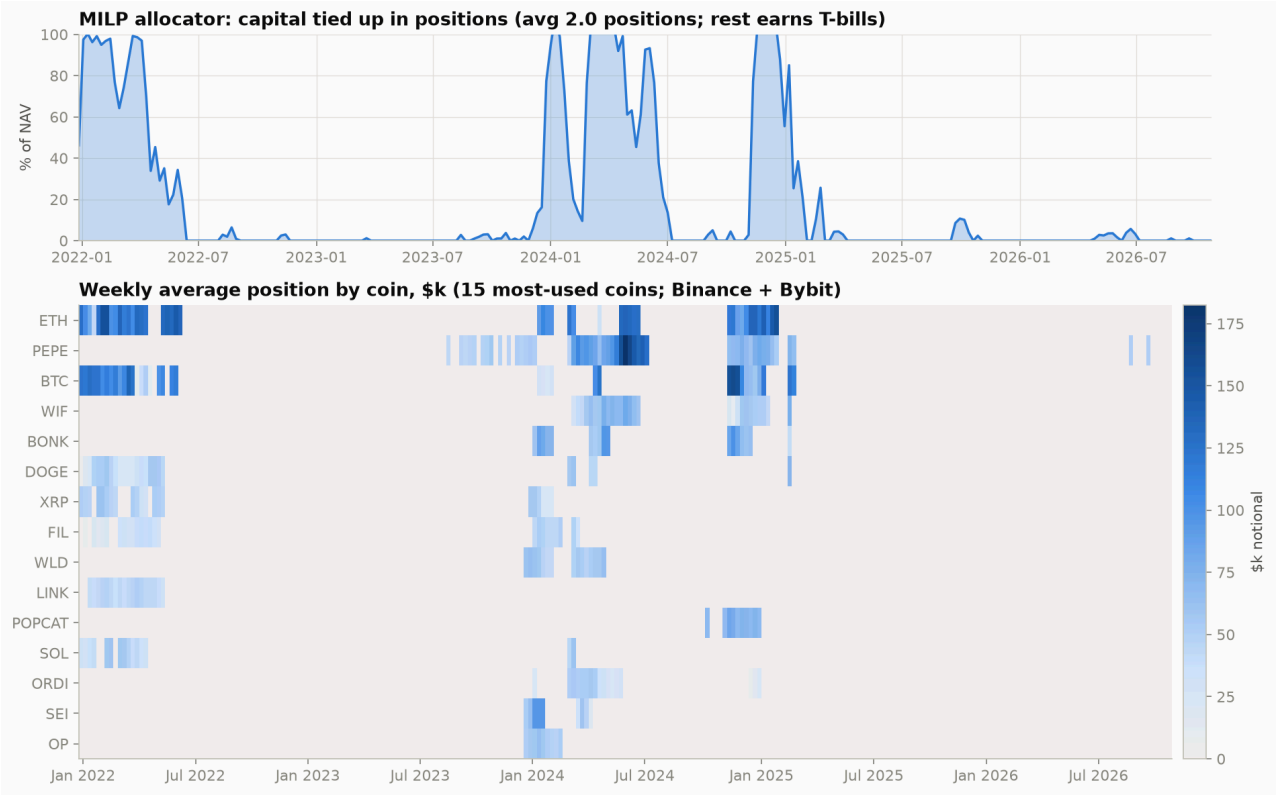


Figure 2. MILP capital deployed and weekly positions by coin. Fully invested in 2022 and the 2024 rallies; mostly in T-bills otherwise.

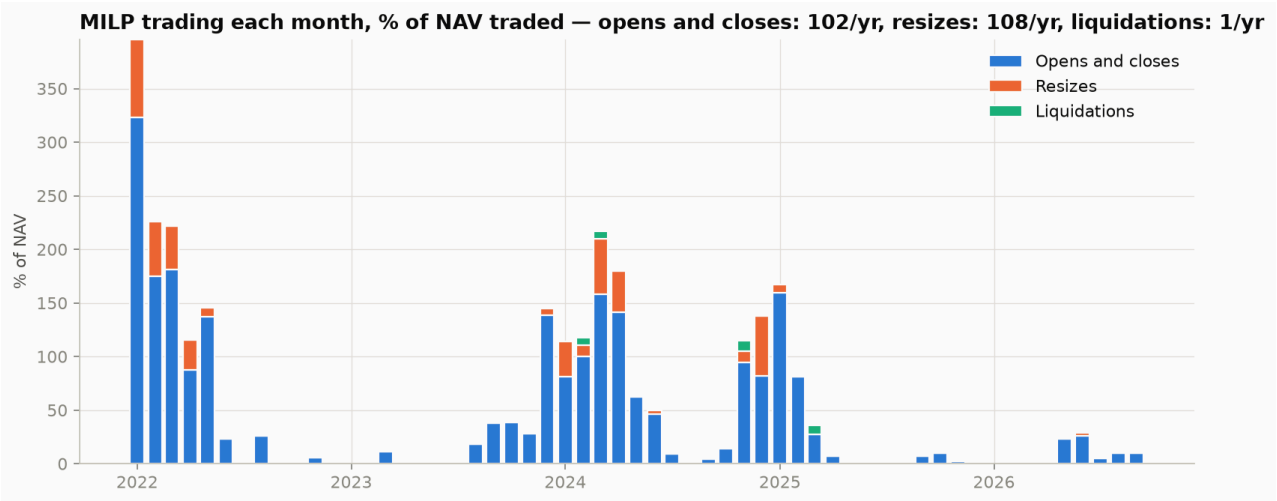


Figure 3. MILP trading by month: opens and closes dominate, concentrated in the high-funding periods.

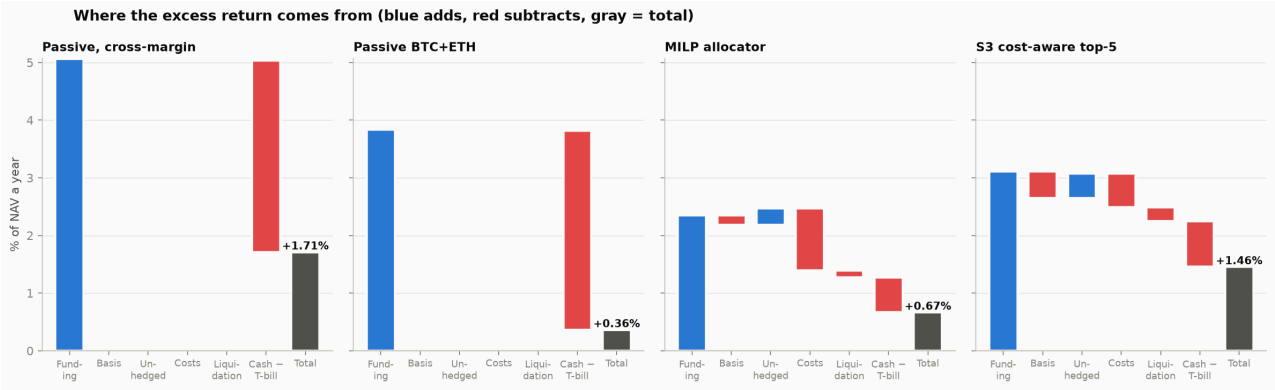


Figure 4. Annual P&L by source. Funding adds (blue); the largest drag in every strategy is "cash − T-bill", the interest given up on capital tied up in positions.

4. What is left: single-exchange basis carry

With selection and arbitrage ruled out, the return is the carry itself, and how much of it survived 2021 depends on the regime:

Year	Context	BTC funding (% a year)	Median coin funding	T-bill	Funding prints negative
2021	Bull market, retail leverage	30.6	10.95	0.05	8%
2022	LUNA, 3AC/Celsius, FTX collapses	4.2	2.9	2.1	42%
2023	Recovery; policy rate about 5%	7.9	10.95	5.3	19%
2024	Spot BTC ETFs, rally	11.9	10.95	5.2	13%
2025	Weaker leverage demand	5.1	6.75	4.2	29%
2026 (to Oct)	BTC mostly below the floor	3.1	4.8	3.8	33%

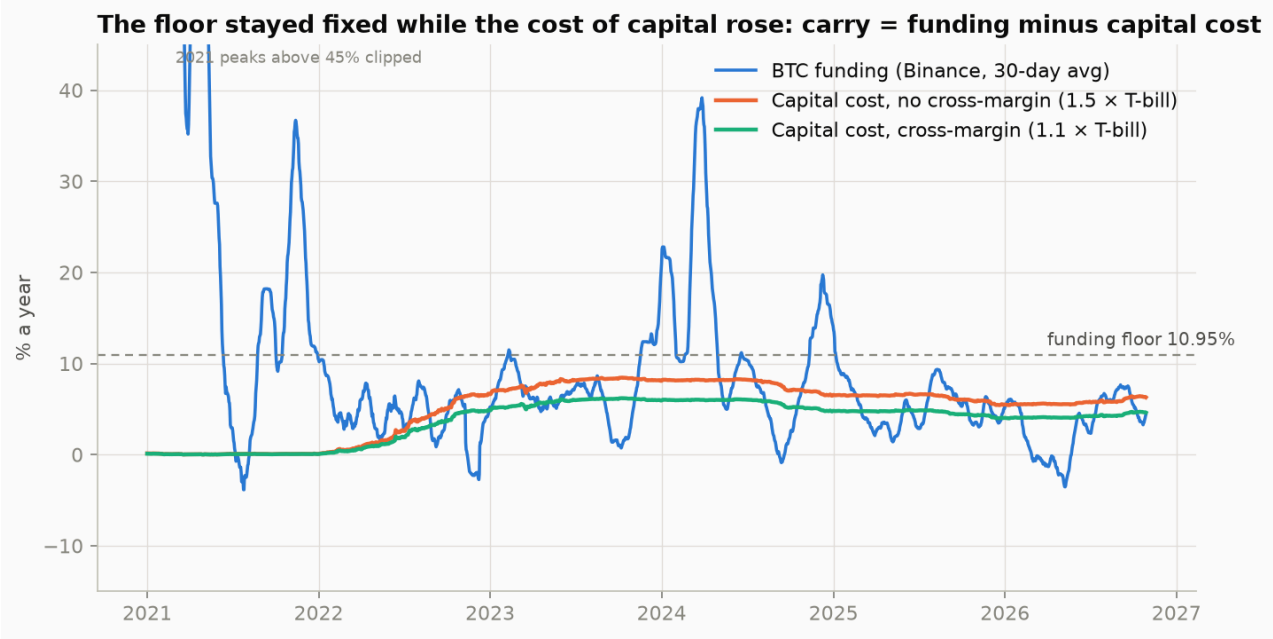


Figure 5. BTC funding (30-day average) against the cost of capital. The trade earns only when funding is above the capital-cost line: continuously in 2021, in bursts since.

Why carry compressed:

- **The floor is set by the exchange, not by interest rates.** 10.95% a year was 11% over cash in 2021 and about 6% over cash once T-bills reached 5%; about 2.5% after the capital cost at margin factor 1.5.
- **Leverage demand is cyclical:** 30.6% BTC funding in 2021, negative 42% of the time in 2022.
- **The perp has traded below spot since 2022** (median −2 to −6 bp): more natural short sellers than leveraged buyers.
- **More arbitrage capital** (spot ETFs and Ethena's USDe stablecoin in 2024) is a likely contributor, not tested here.
- **Fewer coins can be hedged:** new coins increasingly list as perps only, so spot is available for only 72–78% of the top 30 by 2026, down from 100%.

Passive BTC+ETH carry, 2022–2026:

Margin factor	Excess return a year	Sharpe
1.50 (separate accounts)	+0.36%	1.10
1.10	+1.71%	3.85
1.05 (portfolio margin)	+1.94%	4.18

In 2021 alone, before any of the tests above, the same position earned +12.6% over T-bills.

5. Sensitivity to margin, leverage and costs

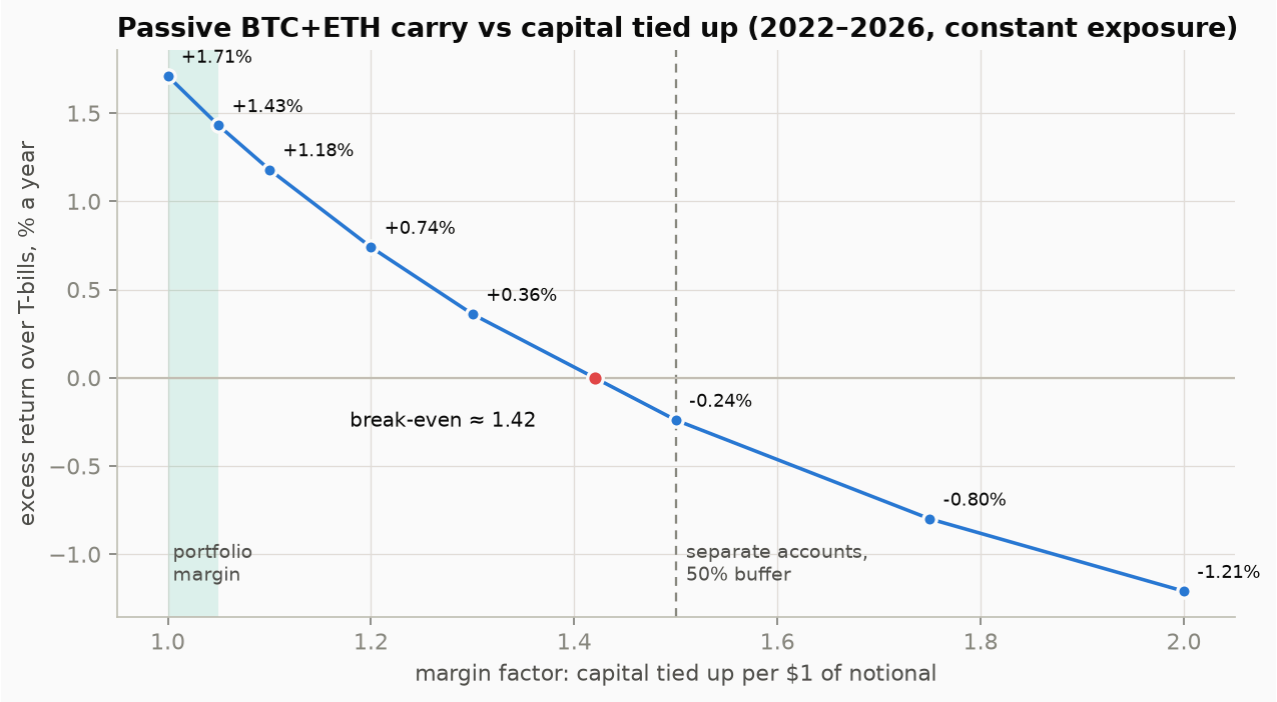


Figure 6. Passive BTC+ETH excess return against capital tied up per $1 of notional (constant exposure, weekly rebalance). Break-even is about 1.4.

Margin is the main lever.

- Each additional 0.1 of margin factor costs about 0.4–0.5% a year.
- Break-even ≈ funding ÷ (T-bill + costs) ≈ 6.4% ÷ 4.6% ≈ **1.4**.
- Exchange documentation puts realistic BTC/ETH portfolio margin at 1.0–1.05 (95% collateral ratio, 0.4% maintenance margin).

Leverage helps little. Borrowing to hold more notional (margin factor 1.05):

Leverage	1×	1.5×	2×	3×
Borrow at T-bill + 1%: excess / max drawdown	+1.41% / −2.1%	+1.65% / −3.4%	+1.86% / −5.2%	+2.28% / −8.5%
Borrow at 11% (exchange loan): excess / max drawdown	+1.28% / −2.2%	−1.31% / −8.2%	−4.05% / −17.8%	−9.55% / −36.8%

- Leverage pays only if funding beats the borrowing rate plus costs. At cash-like rates each extra 1× adds about 0.4% a year but roughly doubles the drawdown; at exchange loan rates it is destructive.
- Better than leverage: **earn interest on the capital already tied up** (yield-bearing collateral), which attacks the largest cost without adding risk.

Costs matter most for active strategies. All fees and slippage multiplied by k (k = 1 is the base case, about 0.38% per round trip):

k	0.25	0.5	1	2
Passive BTC+ETH (margin 1.10)	+1.56%	+1.43%	+1.18%	+0.68%
MILP (margin 1.50)	+1.06%	+0.82%	+0.67%	+0.31%

Even with maker or VIP fees, the MILP stays below passive under the same margin.

6. Conclusion

- The carry that survived 2021 is a **financing spread**: funding near an exchange-set floor, earned against the cost of cash.
- It cannot be improved by arbitrage between the largest exchanges (already integrated), by coin selection (the edge is about the cost of trading it), or by a constrained optimiser (it gives the carry back in turnover).
- What moves returns is the balance sheet: **the margin factor, whether collateral earns interest, and execution cost**.
- Where moving capital is harder (on-chain exchanges), wider gaps persist, consistent with the cost of arbitrage capital being the binding constraint.

Limitations

- Five out-of-sample years, with most of the P&L in 2024. Low-volatility carry flatters the Sharpe ratio: exchange failure, stablecoin de-pegs and liquidation cascades barely appear in five years of daily data.
- 8-hour prices only; flat base-tier fees; an approximate liquidation rule; margin modelled as a single factor.
- No historical coin-borrow data, so the reverse trade (long perp, short spot) is untested.
- Neither Binance nor Bybit serves Ontario residents: this describes the market, not an accessible strategy.

Appendix A. Data and universe

- **Sources:** Binance USD-M API and bulk archive (data.binance.vision), used for delisted symbols the live API no longer serves; Bybit v5 API (spot only from July 2021); Hyperliquid and dYdX public APIs (hourly funding); FRED for the T-bill, lagged one day.
- **Universe:** monthly top 30 by trailing 30-day Binance perp volume among coins traded on both exchanges on at least 25 of the previous 30 days. Stablecoins excluded. Stock, ETF and commodity perps (Binance tag `TRADIFI_PERPETUAL`) excluded; by 2026 they held about a third of top-30 volume.

- **Common grid:** exchanges settle some coins every 1 or 4 hours, so all funding is summed into 8-hour windows ending 00:00, 08:00 and 16:00 UTC. Prices are 8-hour closes; perp prices are divided by contract multipliers (e.g. 1000PEPE).
- **Investable:** spot market on the same exchange, and a perp–spot gap within ±20% (larger gaps are collapses or broken pairs, e.g. LUNA in May 2022). Under 0.5% of windows are missing.
- **On-chain comparison:** Hyperliquid has the same 0.01% floor as Binance (median coin exactly on it), so its gaps are comparable. dYdX's funding has no floor and no dead zone (median 0.00% a year), so small price moves pass straight into it while Binance's stays flat: correlation 0.09 and largely mechanical gaps. Each on-chain pair is compared with Binance–Bybit on the same coins and windows.

Appendix B. Research protocol

Pre-registration. Each test's design and pass/fail rule was written down before it ran:

Test	Rule	Outcome
Cross-exchange arbitrage	Top-20% gap beats 21 bp on average and in ≥ 4 of 6 years	Fail (3 of 6)
Forecast model choice	Regression used only if it beats last value and EWMA on error in ≥ 4 of 5 years	Pass (4 of 5)
Forecast-driven selection	Cost-aware top 5 beats passive Sharpe ($P < 0.10$) and is positive in ≥ 3 of 5 years	Fail ($P = 0.51$)
MILP	Beats passive Sharpe ($P < 0.10$) and positive in ≥ 3 of 5 years	Fail ($P = 0.63$)
MILP with turnover controls	Beats passive at the same margin ($P < 0.10$), positive in ≥ 3 of 5 years	Fail ($P = 0.99$)
On-chain gaps	Both on-chain pairs show wider and slower-closing gaps in ≥ 2 of 3 years	Pass

Statistics used:

- **Paired block bootstrap:** resamples blocks of days from two strategies together, to test whether one Sharpe ratio is really higher than another.
- **Deflated Sharpe ratio:** the probability that a strategy's Sharpe is real after accounting for how many variants were tried (16 here). The best of many skill-free strategies looks good by luck; this measure asks whether a result beats that luck benchmark. Only passive BTC+ETH under portfolio margin survives.
- **Rank IC:** at each point in time, the correlation between how coins are ranked by forecast and how they rank by actual funding afterwards, averaged over time. Zero is no information; above 0.05 is useful in equity research; the forecasts here reach about 0.5 because funding is persistent.

Forecast details:

- Features: last funding; EWMAs with 1, 7 and 30-day half-lives; perp–spot gap; share of the last 3 days at the floor.

- The EWMA model's half-life is chosen from {1, 3, 7, 30} days at each monthly refit, using training data only.
- No look-ahead: a training row is used only if its whole target window ends before the refit date. A test scrambles all data after a cut-off and checks that no earlier forecast changes.

Appendix C. Backtest engine and assumptions

- **Timing:** decide at an 8-hour close using data up to then; trade at the next close.
- **P&L per window:** funding + basis change + cash interest − costs − liquidation penalties. The components add up to the equity change exactly (tested).
- **Costs (base tier, taker):** spot 0.10%; perp 0.05% (Binance), 0.055% (Bybit); plus 2 bp slippage per leg: about 0.38% per round trip.
- **Margin:** 50% buffer on the short, no cross-margining (margin factor 1.5) unless stated; margin topped up once a day.
- **Liquidation:** if the perp rises more than 40% since the last daily top-up, the short is closed with a 2% penalty and the spot is sold at the next close.
- **Exits:** coins that stop being investable are closed with full costs; coins with no price for 3 days (delisted) are closed at their last prices.
- **Liquidity:** positions capped at 1% of 7-day average daily volume.

Appendix D. MILP formulation

For each candidate coin i on exchange v, with current notional x^0 and forecast annual funding f:

- **Variables:** target notional $x \geq 0$; buys $b \geq 0$; sells $s \geq 0$ ($s \leq x^0$); open position $y \in \{0, 1\}$; for each direction between the two exchanges, transfer amount $g \geq 0$ and transfer flag $z \in \{0, 1\}$.
- **Objective (maximise, over a horizon H of 30 days):** $\Sigma\, x \cdot (f - m \cdot r - \text{penalty}) \cdot H - \Sigma\, b \cdot$ round-trip cost $- \Sigma\, g \cdot r / 365 - \$10 \times \Sigma\, z$, where m is the margin factor, r the T-bill rate, and the penalty is 0% (BTC/ETH), 3% (other coins) or 10% (coins listed under 90 days) a year.
- **Constraints:**
- $x = x^0 + b - s$;
- $x \leq \text{cap} \cdot y$ and $x \geq \$10\text{k} \cdot y$, where cap = min(per-coin share of exchange capital, 1% of daily volume);
- $\Sigma\, y \leq 12$;
- $\Sigma\, m \cdot x$ on each exchange ≤ its capital − transfers out + transfers arriving;
- each exchange ≤ 60% of total capital after transfers;
- $\$50\text{k} \cdot z \leq g \leq$ total capital $\cdot z$.
- **Turnover-controlled version:** sells also pay the exit half of the round trip, and each held position either stays unchanged or trades by at least max($5k, 10% of the position).
- Solved daily with HiGHS (via `scipy.optimize.milp`); positions left unchanged are held as is rather than re-traded as prices drift.

Appendix E. Corrections

Errors found after first results were fixed, tested and rerun; no decision changed.

Found by	Problem	Effect of the fix
Engine refactor	A coin with spot but no perp price produced undefined equity, silently dropping months of one variant	EWMA variant Sharpe 2.34 → 1.35
30-day forecasts	Look-ahead gap hard-coded to 7 days	Prevented a leak before any 30-day result
Position charts	Delisted coins (e.g. MATIC after its rename) held forever with no price	Active strategies improved
Trade counts	Optimiser positions re-traded as prices drifted	MILP resizes fell from ~615 to ~105 a year
Data end date	Final window lacked a funding print, forcing an artificial exit	Passive Sharpe 1.01 → 1.10

Appendix F. Reproducing the results

```
python -m venv .venv && .venv/bin/pip install -e ".[dev]"
.venv/bin/python research/build_data.py              # data (~1.5 h the first time;
resumable)
.venv/bin/python research/phase1_descriptives.py     # cross-exchange test and
descriptives
.venv/bin/python research/phase2_forecast.py         # investable set, forecasts
.venv/bin/python research/phase3_backtest.py         # coin-selection rules
.venv/bin/python research/phase4_allocator.py        # MILP
.venv/bin/python research/phase4b_allocator.py       # MILP with turnover controls
.venv/bin/python research/sensitivity.py             # margin, leverage, costs
.venv/bin/python research/build_extensions.py hyperliquid && .venv/bin/python
research/extension_a_onchain.py
.venv/bin/python research/figures.py                 # figures
.venv/bin/pytest                                     # tests
```

Detailed write-ups per stage: `docs/phase1.md` to `docs/phase4b.md`, `docs/sensitivity.md`, `docs/extension_a.md`. Decision rules: `docs/plan.md`. Every decision and correction: `docs/decisions.md`.